

Notes: Flynn, Ghent, and Tchisty (RFS, 2020)

Informational Efficiency in Securitization after Dodd-Frank

Contents

1	Big picture	3
2	Institutional detail and data (key objects)	4
2.1	The Dodd–Frank risk-retention rule	4
2.2	Three retention structures	4
2.3	Simple example: why horizontal retention is much riskier	4
2.4	The B-piece buyer	5
2.5	Data sources and sample	5
2.6	Key measured variables	5
3	Retention stringency and signaling constraints	6
3.1	The rule is binding	6
3.2	Why binding matters for information	6
3.3	The rule is stringent relative to historical losses	7
3.4	The post-rule collateral looks safer than crisis-era collateral	7
3.5	Main implication of the stringency evidence	7
4	Model and mechanisms (all equations + interpretation)	8
4.1	Basic setup	8
4.2	Mandated retention choices	8
4.3	Issuer impatience and the cost of holding risk	9
4.4	Issuer payoffs	9
4.5	Why horizontal retention signals quality	10
4.6	The partially separating equilibrium	10
4.7	Uniform-distribution example	11
4.8	Difference from DeMarzo and Duffie (1999)	11
5	Empirical specifications and identification	11
5.1	Baseline pricing regression	11
5.2	Heterogeneity by tranche information sensitivity	12
5.3	Magnitude of asymmetric information	12
5.4	What makes the empirical design credible?	13
5.5	Main limitations	13
6	Tables and figures	14

6.1	Figure 1: Risk retention types	14
6.2	Table 1: Variable definitions	14
6.3	Table 2: Security-level summary statistics	15
6.4	Table 3: Risk retention amount	16
6.5	Figure 2: Losses and B-piece size	16
6.6	Figure 3: Density of B-piece size	16
6.7	Figure 4: B-piece size versus cumulative losses	17
6.8	Figure 5: Issuer payoffs	17
6.9	Table 4: Yield spreads and the form of risk retention	17
6.10	Table 5: Heterogeneous effects	17
6.11	Table 6: Robustness	19
6.12	Appendix C tables: counterfactual loss analysis	20
7	Policy implications	20
7.1	Investor sophistication	20
7.2	The rule may reduce informational efficiency	21
7.3	Two positive design features	21
7.4	Dynamic retention policy	21
8	Short summary	22
9	Conclusion	22

1 Big picture

- **Question.** How does Dodd–Frank mandated risk retention affect the information investors can infer from securitizers’ choices in the commercial mortgage-backed securities (CMBS) market?
- **Policy setting.** The Dodd–Frank risk-retention rule requires CMBS sponsors to retain an interest equal to at least **5% of deal value**. The rule became effective for CMBS on December 24, 2016.
- **Main institutional fact.** The 5% requirement is both **binding** and **stringent**. In the deals the authors can observe, issuers do not materially retain more than the required minimum. This means issuers cannot use the *level* of retention to signal quality.
- **Main economic question.** If the regulation fixes the amount of skin in the game, can the market still learn something about collateral quality from the *structure* of skin in the game?
- **Retention structures.** The rule allows three ways to meet the 5% requirement:
 - **Vertical retention:** retain 5% of every tranche.
 - **Horizontal retention:** retain a first-loss residual interest, similar to the B-piece.
 - **Hybrid retention:** combine vertical and horizontal exposure.
- **Key intuition.** Horizontal retention is more exposed to collateral losses than vertical retention. If holding the first-loss piece is costly, only issuers with better private information about collateral quality will be willing to choose it. Thus, horizontal retention can become a costly signal.
- **Model result.** The paper builds a signaling model in which the issuer observes a private component of expected losses and chooses the retention structure after learning this information. There is a partially separating equilibrium: better collateral pools choose horizontal retention, worse collateral pools choose vertical retention.
- **Empirical result.** Securities in horizontally retained deals sell at lower issue spreads than securities in vertical or hybrid deals. The baseline spread reduction is about **5.9 to 11 basis points**.
- **Where the effect is strongest.** The pricing effect is concentrated in the most information-sensitive investment-grade tranches. BBB-rated bonds have spreads about **37 to 75 basis points** lower in horizontal deals, depending on specification. Non-AAA senior tranches have spreads about **10 to 17 basis points** lower.
- **Quantification of asymmetric information.** Combining the spread evidence with the model implies that only about **0.7% to 1.3%** of deal value is attributable to asymmetric information between issuers and investors in the 2017–2019 CMBS market.
- **Policy message.** The rule may protect senior investors from losses, but it also removes a potentially more precise signaling channel: the level of retention. The market still learns from retention structure, but that signal is coarser.
- **Economic takeaway.** Mandated skin in the game does not simply create incentives. It also changes the information environment. A rule that is too stringent can eliminate level-based signaling and leave only structure-based signaling.

2 Institutional detail and data (key objects)

2.1 The Dodd–Frank risk-retention rule

The CMBS risk-retention rule requires sponsors to retain an interest equal to at least 5% of the deal value at issuance. The paper focuses on deals issued after the rule became effective in late 2016.

Interpretation. The regulation is meant to align the incentives of securitizers and investors. The concern is that issuers may know more about loan quality than outside investors and may otherwise sell risky securities without keeping enough downside exposure.

2.2 Three retention structures

1. Vertical retention.

- The issuer retains 5% of the par value of each tranche.
- Because the issuer owns a slice of every security, the retained stake is mechanically 5% of the market value of the deal.
- Loss exposure is proportional across the capital structure.

2. Horizontal retention.

- The issuer retains, or arranges for a qualified third-party buyer to retain, a first-loss position.
- The horizontal residual interest must have market value at least 5% of the market value of all securities issued.
- Because the first-loss B-piece usually trades below par, 5% of market value often corresponds to a much larger percentage of deal par value.
- In the post-rule data, horizontal B-pieces in conduit deals price at roughly **48% to 56% of par**. Thus, a 5% market-value requirement corresponds roughly to about **10% of par**.

3. Hybrid retention.

- The issuer combines a vertical slice and a horizontal first-loss slice.
- The total retained market value must still equal at least 5%.
- Hybrid deals are sometimes called L-shaped retention deals.

2.3 Simple example: why horizontal retention is much riskier

Suppose a deal has \$100 par value of collateral, a \$90 senior security, and \$10 of junior securities. The senior security prices at par, but the junior securities price at 50% of par, so the deal market value is \$95.

- **Vertical retention:** the issuer retains 5% of each tranche. This is \$4.50 of senior par and \$0.50 of junior par, worth \$4.75 in total.
- **Horizontal retention:** the issuer retains \$4.75 of market value entirely in junior securities. Since the junior securities trade at 50% of par, the issuer must retain \$9.50 of junior par.

- **Loss comparison:** if the collateral suffers \$10 of losses with zero recovery, the horizontal holder loses \$9.50, whereas the vertical holder loses only \$0.50.

Interpretation. Horizontal retention concentrates exposure in the first-loss layer. It is therefore a more costly and more informative signal of confidence in collateral quality.

2.4 The B-piece buyer

In practice, the horizontal piece is often sold to a qualified third-party B-piece buyer. This does not destroy the signal for three reasons:

- The B-piece buyer is sophisticated and typically performs extensive due diligence.
- The buyer has access to detailed underwriting information and can influence collateral selection, including requesting that loans be removed from the pool.
- The buyer must retain the interest for at least five years, so the buyer is exposed to default risk rather than immediately reselling the risk.

Interpretation. In a horizontal deal, the B-piece sale price reflects informed expectations about collateral performance. Therefore, the issuer is still affected by the quality of the collateral when the transaction is structured and priced.

2.5 Data sources and sample

The empirical analysis uses post-rule U.S. CMBS issuance from **January 1, 2017 to June 30, 2019**. The main sources are CRE Direct, Trepp, and Bloomberg.

- The authors observe risk-retention data for **297** conduit and large loan/single-borrower CMBS deals.
- The sample includes **114** conduit deals and **183** large loan/single-borrower deals.
- There are **139** horizontal deals, **129** vertical deals, and **29** hybrid deals.
- The security-level sample has **3,332** non-IO securities in the summary statistics.
- The issue-spread regressions use securities for which yield spreads at issue are observed; the summary table reports **1,809** spread observations.
- The average deal size is about **\$730 million**; the average deal contains about **13** tranches.
- About **55%** of securities in the sample are from conduit deals.

Interpretation. The data cover most of the early post-Dodd–Frank CMBS market. The paper is not comparing pre- and post-rule pricing directly; it studies how issuers choose among allowed retention structures under the new rule.

2.6 Key measured variables

- **Spread:** yield spread at issue relative to the relevant benchmark. The average spread in the sample is about **1.38 percentage points**, or 138 basis points.

- **horizrr**: indicator for a pure horizontal retention deal.
- **verticalrr**: indicator for a pure vertical retention deal.
- **hybridrr**: indicator for an L-shaped/hybrid retention deal.
- **origsubpct**: original subordination percentage of the tranche.
- **nonAAAsr**: indicator for senior investment-grade tranches rated below AAA, from AA+ to BBB-.
- **BBB**: indicator for BBB+, BBB, or BBB- tranches.
- **horizrr_c**: continuous measure equal to 1 for pure horizontal, 0 for pure vertical, and the horizontal share of the 5% retention for hybrid deals.

Interpretation. The key empirical object is the pricing difference between securities from horizontal and non-horizontal deals, conditional on rating, subordination, deal type, collateral observables, and fixed effects.

3 Retention stringency and signaling constraints

3.1 The rule is binding

The authors manually inspect deal documents and combine them with CRE Direct data to determine whether issuers retain materially more than the legal minimum.

- In pure vertical deals, the retained par value is always exactly **5%**.
- In horizontal deals with available documents, the fair value of the horizontal piece is between **4.99% and 5.22%** of deal market value.
- In hybrid deals with available information, total retention ranges from **5% to 5.15%**.

Interpretation. Issuers choose the minimum or a tiny amount above the minimum. Small excesses are likely compliance buffers, not meaningful voluntary retention.

3.2 Why binding matters for information

Classic signaling models such as Leland and Pyle (1977), DeMarzo and Duffie (1999), and DeMarzo (2005) emphasize signaling through the *quantity* retained. If better issuers voluntarily retain more, investors can infer quality from the level of retention.

In this paper’s setting, that mechanism is shut down:

- the regulation requires a high minimum;
- the minimum binds in practice;
- issuers do not choose materially higher retention levels.

Interpretation. The market cannot infer fine-grained quality from how much the issuer retains. The only remaining choice is *how* the issuer retains the required exposure.

3.3 The rule is stringent relative to historical losses

The paper evaluates stringency by comparing the 5% market-value requirement to historical CMBS losses and B-piece sizes.

The key conversion is:

$$5\% \text{ of market value} \approx 10\% \text{ of par value}, \quad (1)$$

because the first-loss horizontal piece often trades at about 50% of par.

Using 2000–2016 vintage conduit CMBS, the paper documents:

- cumulative principal losses are below 10% of original deal balance in all vintage years except 2008;
- the 2008 vintage has only 9 deals, compared with an average of 74 deals per year before the rule;
- B-piece sizes were usually below 5% of par during 2000–2016;
- the B-piece exceeded 10% of par in less than 1% of 2000–2016 conduit deals.

Interpretation. A 5% market-value horizontal requirement is large relative to the loss protection that historical conduit CMBS investors usually needed. For most historical vintages, the required horizontal position would have been far larger than necessary to protect senior bondholders from principal losses.

3.4 The post-rule collateral looks safer than crisis-era collateral

The authors also conduct a counterfactual loan-level loss analysis. They estimate loss probabilities and loss magnitudes using 2006–2008 loans, then ask how 2017–2019 loans would have performed if they had faced the same crisis-era mapping from observables to losses.

Key comparisons:

- 2017–2019 loans have lower LTV than 2006–2008 loans: about **59.4** versus **68.65**.
- 2017–2019 loans have higher NOI-based DSCR: about **2.31** versus **1.67**.
- Actual 2006–2008 loans had about a **19.5%** probability of any loss; predicted probability for 2017–2019 loans is roughly **13%**.
- Actual 2006–2008 loans had about a **16.2%** probability of a loss above 10%; predicted probability for 2017–2019 loans is roughly **10%**.
- Actual mean loss for 2006–2008 loans is about **9%** of face value; predicted mean loss for 2017–2019 loans is about **5.5% to 6.2%**.

Interpretation. Even under a crisis-style counterfactual, post-rule collateral would likely have generated losses below the effective 10% par threshold implied by a 5% market-value horizontal piece.

3.5 Main implication of the stringency evidence

The paper’s first empirical contribution is not the spread regression; it is the institutional fact that the retention requirement binds. This creates the puzzle and motivates the model.

- If retention were not binding, issuers could signal quality by voluntarily retaining more.
- Since retention is binding, level-based signaling disappears.
- Since issuers still choose among vertical, horizontal, and hybrid structures, structure-based signaling may remain.

Economic intuition. A very high required first-loss position can make extra retention uninformative. If the required piece is already large enough to absorb nearly all plausible losses, retaining more does not meaningfully add risk exposure.

4 Model and mechanisms (all equations + interpretation)

4.1 Basic setup

The model has a risk-neutral issuer and risk-neutral outside investors. The issuer owns a pool of commercial mortgages. The pool's maximum possible cash flow is normalized to 1.

The realized cash flow is

$$A = 1 - \alpha - \beta. \quad (1)$$

Objects.

- A : realized payoff of the mortgage pool.
- α : loss component known only by distribution to both issuer and investors.
- β : loss component privately observed by the issuer.
- $\alpha_0 = \mathbb{E}[\alpha]$ and $\beta_0 = \mathbb{E}[\beta]$.
- $\bar{\alpha}$ and $\bar{\beta}$: upper bounds of the support of α and β .

Interpretation. The variable β captures soft information about loan quality. A high β means higher expected loss and lower collateral quality. Investors know the distribution of β , but not its realization.

4.2 Mandated retention choices

The issuer must retain a fraction

$$v = 0.05 \quad (2)$$

of the deal value.

There are two retention choices in the model:

- **Vertical retention:** the issuer retains a fraction v of the whole pool.
- **Horizontal retention:** the issuer retains a first-loss tranche with principal balance h , where the tranche's market value equals v of total deal value.

With vertical retention, the retained payoff is

$$vA, \quad (3)$$

and the sold payoff is

$$(1 - v)A. \quad (4)$$

With horizontal retention, investors receive

$$\min(A, 1 - h), \quad (5)$$

and the issuer receives

$$\max(0, A - (1 - h)). \quad (6)$$

Interpretation. Vertical retention spreads exposure across the whole deal. Horizontal retention concentrates exposure in the residual after the senior securities are paid.

4.3 Issuer impatience and the cost of holding risk

The issuer wants to sell the mortgage pool because it is financially constrained, faces capital requirements, or values immediate cash more than future cash flows. The paper models this by assuming the issuer discounts retained cash flows more heavily than outside investors.

Let

$$\delta_h < \delta_v < 1. \quad (7)$$

- δ_v : issuer's valuation discount factor for the vertical retained piece.
- δ_h : issuer's valuation discount factor for the horizontal first-loss piece.

Interpretation. The first-loss piece is more costly to hold because it ties up more capital and is harder to finance. Therefore δ_h is lower than δ_v .

4.4 Issuer payoffs

Let P_v and P_h denote the market values of the sold securities under vertical and horizontal retention. These prices cannot depend directly on the issuer's private realization of β , because outside investors do not observe it.

The issuer's expected payoffs are

$$\pi_v(\beta) = P_v + \delta_v v \mathbb{E}[A \mid \beta], \quad (8)$$

$$\pi_h(\beta) = P_h + \delta_h \mathbb{E}[\max(0, A - (1 - h)) \mid \beta]. \quad (9)$$

Since

$$\mathbb{E}[A \mid \beta] = 1 - \alpha_0 - \beta, \quad (10)$$

the vertical payoff is

$$\pi_v(\beta) = P_v + \delta_v v (1 - \alpha_0 - \beta). \quad (2)$$

The paper uses the stringency result to assume that the horizontal piece is larger than all possible realized losses:

$$h > \bar{\alpha} + \bar{\beta}. \quad (11)$$

Then the horizontal retained payoff simplifies to

$$\mathbb{E}[\max(0, A - (1 - h)) \mid \beta] = h - \alpha_0 - \beta, \quad (12)$$

and

$$\pi_h(\beta) = P_h + \delta_h(h - \alpha_0 - \beta). \quad (3)$$

Interpretation. The horizontal payoff is more sensitive to β than the vertical payoff. If $\delta_v v < \delta_h$, then the slope of $\pi_h(\beta)$ is more negative than the slope of $\pi_v(\beta)$. This means bad private information hurts the horizontal payoff more.

4.5 Why horizontal retention signals quality

A high-quality issuer has low β . Since horizontal retention puts the issuer in the first-loss position, choosing horizontal retention is attractive only when the issuer knows the pool is relatively safe.

A low-quality issuer has high β . Because the first-loss payoff is very sensitive to losses, a high- β issuer prefers the less sensitive vertical structure.

Core mechanism.

$$\begin{aligned} &\text{horizontal is costly} + \text{cost depends on private quality} \\ &\implies \text{horizontal can signal high quality.} \end{aligned} \quad (13)$$

4.6 The partially separating equilibrium

Theorem 1 shows that under a regularity condition, there exists a cutoff $\hat{\beta}$ such that:

$$\beta \leq \hat{\beta} \implies \text{choose horizontal retention,} \quad (14)$$

$$\beta > \hat{\beta} \implies \text{choose vertical retention.} \quad (15)$$

The equilibrium horizontal principal balance is

$$h = v + (1 - v)(\alpha_0 + \beta_h), \quad (5)$$

where

$$\beta_h = \mathbb{E}[\beta \mid \beta \leq \hat{\beta}], \quad \beta_v = \mathbb{E}[\beta \mid \beta > \hat{\beta}]. \quad (16)$$

The market values of sold securities are

$$P_v = (1 - v)(1 - \alpha_0 - \beta_v), \quad (6)$$

$$P_h = 1 - h. \quad (7)$$

Interpretation. Since $\beta_h < \beta_v$, horizontal deals have lower expected losses than vertical deals. Investors understand this and price the sold securities in horizontal deals more favorably.

4.7 Uniform-distribution example

If β is uniformly distributed, the model has a closed-form solution. In the numerical example, the authors set:

$$v = 0.05, \quad \beta_0 = 0.005, \quad \alpha_0 = 0.05, \quad (17)$$

$$\delta_v = 0.98, \quad \delta_h = 0.92. \quad (18)$$

This implies that asymmetric information accounts for 1% of pool value, while total expected losses are 5.5% of principal.

The equilibrium is:

$$\hat{\beta} = 0.00438, \quad h = 0.0996, \quad (19)$$

$$P_v = 0.8954, \quad P_h = 0.9004. \quad (20)$$

Interpretation.

- The horizontal piece is about **9.96% of total CMBS principal**, consistent with a first-loss piece priced at about half of par.
- Sold securities in horizontal deals have higher value: $P_h > P_v$.
- About **44%** of issuers choose horizontal retention in the example.
- The horizontal payoff has a much steeper slope in β than the vertical payoff, which makes the retention structure informative.

4.8 Difference from DeMarzo and Duffie (1999)

The model is related to DeMarzo and Duffie, but the timing is different.

- In DeMarzo and Duffie, security design is chosen before the issuer learns private information. Therefore the security design itself cannot signal quality.
- In this paper, the issuer chooses vertical or horizontal retention after learning private information. Therefore the security design choice can be a signal.

Interpretation. The regulation changes the signaling margin. Quantity signaling is constrained by the 5% rule, so the model shifts the action to structure signaling.

5 Empirical specifications and identification

5.1 Baseline pricing regression

The baseline empirical specification is

$$spread_{i,j,t} = \beta_0 + \beta_1 horizrr_{j,t} + \beta_2 hybridrr_{j,t} + \beta_x Cont_{i,j,t} + \varepsilon_{i,j,t}. \quad (11)$$

Objects.

- $spread_{i,j,t}$: issue spread to benchmark for security i in deal j issued in quarter t .

- $horizrr_{j,t}$: indicator for pure horizontal retention.
- $hybridrr_{j,t}$: indicator for hybrid retention.
- $Cont_{i,j,t}$: security-level and deal-level controls.

Controls. The controls include broad rating indicators, subordination, tranche weighted average life, deal size, number of securities, collateral maturity, collateral DSCR, collateral LTV, collateral coupon, tranche balance, lead-manager volume, depository-lead-manager indicator, floating-rate indicator, initial coupon, deal type fixed effects, quarter fixed effects, property-type share controls, and in some specifications lead-manager and B-piece-buyer fixed effects.

Interpretation. A lower spread means a higher issue price. If horizontal retention credibly signals better collateral, then β_1 should be negative.

5.2 Heterogeneity by tranche information sensitivity

The model predicts better collateral quality at the deal level, but not every tranche should react equally. AAA tranches are highly protected by subordination, while BBB tranches are closer to the loss boundary.

The paper estimates

$$\begin{aligned} spread_{i,j,t} = & \beta_0 + \beta_1 horizrr_{j,t} + \beta_2 RtgIndic_{i,j,t} \\ & + \beta_3 horizrr_{j,t} \times RtgIndic_{i,j,t} + \beta_x Cont_{i,j,t} + \varepsilon_{i,j,t}. \end{aligned} \quad (12)$$

$RtgIndic$ is either:

- an indicator for BBB-rated tranches; or
- an indicator for non-AAA senior investment-grade tranches, from AA+ to BBB-.

Prediction. $\beta_3 < 0$: horizontal retention should lower spreads more for more information-sensitive tranches.

5.3 Magnitude of asymmetric information

The model can be calibrated using the difference in spreads between horizontal and vertical deals. Assume β is uniformly distributed on $[0, \bar{\beta}]$. The average asset values for horizontal and vertical deals are

$$A_h = 1 - \alpha_0 - \hat{\beta}/2, \quad (13)$$

$$A_v = 1 - \alpha_0 - (\hat{\beta} + \bar{\beta})/2. \quad (14)$$

Therefore,

$$\Delta A \equiv A_h - A_v = \frac{\bar{\beta}}{2}. \quad (21)$$

If Δy is the yield difference between horizontal and vertical deals and D is modified duration,

$$\Delta A \approx -D\Delta y, \quad (22)$$

so

$$\bar{\beta} \approx -2D\Delta y. \quad (15)$$

Using $D = 5.9$ years and the Table 4 spread reduction of 0.059% to 0.11%, the implied information asymmetry is

$$\bar{\beta} \approx 0.7\% \text{ to } 1.3\% \text{ of deal value.} \quad (23)$$

Interpretation. The post-crisis CMBS market appears to have low asymmetric information. Investors are not missing a large hidden component of collateral quality.

5.4 What makes the empirical design credible?

- **Direct institutional variation.** The rule fixes the retention amount but leaves the structure choice open, creating meaningful variation across horizontal, vertical, and hybrid structures.
- **Security-level pricing.** The dependent variable is the issue spread, not just coupon spreads or secondary market prices.
- **Rich controls.** The regressions compare securities with controls for ratings, subordination, deal type, collateral characteristics, lead-manager characteristics, and timing.
- **Tranche-level cross-sectional prediction.** The strongest effects appear for BBB and non-AAA senior securities, exactly where the signal should matter most.
- **Fixed effects.** Some specifications include lead-manager fixed effects and B-piece-buyer fixed effects, reducing concern that the result is only driven by repeated differences across sponsors or buyers.
- **Robustness to hybrid intensity.** A continuous measure of horizontal intensity is negatively related to spreads, supporting the interpretation that more first-loss exposure sends a stronger signal.

5.5 Main limitations

- **Not random assignment.** Issuers choose retention structures, so the spread result should be interpreted as evidence consistent with a signaling equilibrium, not as a randomized causal effect.
- **Short post-rule window.** The main pricing sample covers January 2017 to June 2019. This is natural because the rule is new, but the time window is short.
- **Limited realized-performance evidence.** The post-2017 deals are too recent to observe full credit performance, so the paper relies on pricing and counterfactual loss analysis.
- **CMBS-specific institutional structure.** The B-piece buyer is unusually sophisticated and involved in CMBS. The same conclusions may not apply mechanically to all asset-backed securities.
- **Regulatory objective ambiguity.** The paper evaluates informational efficiency and investor protection, but the 5% rule may also have systemic-risk objectives that are not fully modeled.

6 Tables and figures

6.1 Figure 1: Risk retention types

Read: The figure visually contrasts vertical, horizontal, and hybrid retention.

- Vertical retention is a thin slice of every tranche.
- Horizontal retention is a first-loss position at the bottom of the capital stack.
- Hybrid retention combines a vertical slice with a first-loss slice.

Why it matters. This figure is the institutional foundation of the paper. The same 5% market-value requirement creates very different exposure to collateral losses depending on retention structure.

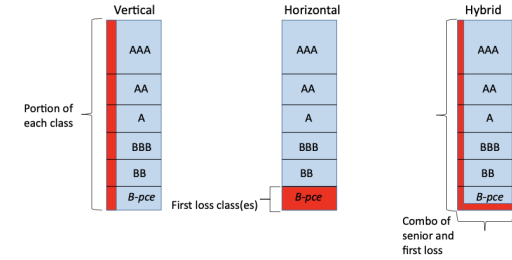


Figure 1
Risk retention types

Table 1

Variable definitions

Variable	Description
A	Indicator equal to 1 if the tranche is rated A+, A, or A- at issue and 0 otherwise
AA	Indicator equal to 1 if the tranche is rated AA+, AA, or AA- at issue and 0 otherwise
AAA	Indicator equal to 1 if the tranche is rated AAA at issue and 0 otherwise
AAA_h	Interaction between (<i>horizrr</i>) and AAA
anyloss	Indicator equal to 1 if the loan has experienced any losses as of December 2017 and 0 otherwise
BB	Indicator equal to 1 if the tranche is rated BB+, BB, or BB- at issue and 0 otherwise
BBB	Indicator equal to 1 if the tranche is rated BBB+, BBB, or BBB- at issue and 0 otherwise
BBB_h	Interaction between (<i>horizrr</i>) and BBB
conduit	Indicator equal to 1 if deal is Conduit and 0 otherwise
cutoffbalance	Total collateral balance of the deal at origination in billions of dollars
cutoffdiscr_ncf	Deal-weighted average debt service coverage ratio at origination based on net cash flow
cutoffdiscr_noi	Deal-weighted average debt service coverage ratio at origination based on NOI
cutofflty	Deal-weighted average LTV at origination
depository	Indicator equal to 1 if the lead manager is a depository institution and 0 otherwise
face	Loan balance at securitization in millions of dollars
firstcoupon	Tranche initial coupon in percentage points
floaterr	Indicator equal to 1 if the tranche is floating rate and 0 otherwise
horizrr	Indicator equal to 1 if the deal is structured with horizontal retention and 0 otherwise
horizrr_c	Variable equal to 1 if (<i>horizrr</i>)=1, 0 if (<i>verticalrr</i>)=1, and the proportion of 5% comprised of horizontal if <i>hybridrr</i> =1
horsize_hrr	Size of horizontally retained piece in a horizontal deal as a percentage of par value
horsize_hyrr	Size of horizontally retained piece in a hybrid deal as a percentage of par value
horsizeofpar	Size of horizontally retained piece as a percentage of par value
hybridrr	Indicator equal to 1 if the deal is structured with L-shaped retention and 0 otherwise
leadmgrsize	Total value of deals originated by lead manager in a given year in billions of dollars
loanhlyrr	Indicator equal to 1 if loan is in deal structured with horizontal or hybrid retention and 0 otherwise
loanhrr	Indicator equal to 1 if loan is in deal structured with horizontal retention and 0 otherwise
lossoverten	Indicator equal to 1 if the loan has experienced losses greater than 10% of face value as of December 2017 and 0 otherwise
lossrate_face	Loan realized loss rate, computed as realized loss divided by loan balance at securitization
mtg_orig_wal	Tranche-weighted average life at origination
nonAAAsr	Indicator equal to 1 if the tranche is rated between AA+ and BBB- at issue and 0 otherwise
nonAAAsr_h	Interaction between <i>horizrr</i> and <i>nonAAAsr</i>
ntranches	Number of tranches in the deal
orig_discrnoi	Loan debt service coverage ratio at origination
orig_wam	Deal-weighted average maturity at origination
origsubpct	Subordination level of tranche expressed as a percentage at origination
rtg_sub	The interaction of a rating indicator (AAA, AA, A, or BBB) with <i>origsubpct</i>
securltv	Loan LTV at time of securitization
securocc	Loan property occupancy rate at securitization
securwac	Loan gross coupon rate at securitization
securwac	Deal-weighted average coupon
spread	Tranche spread to benchmark at time of initial sale in percentage points
term_secur	Loan term to maturity at securitization
verticalrr	Indicator equal to 1 if the deal is structured with vertical retention and 0 otherwise
vertsize	Size of vertically retained piece as a percentage of par value
vertsize_hyrr	Size of vertically retained piece in a hybrid deal as a percentage of par value
vertsize_vrr	Size of vertically retained piece in a vertical deal as a percentage of par value

6.2 Table 1: Variable definitions

Read: The table defines the paper's security-level, deal-level, retention, and counterfactual-loss variables.

Important variables:

- *horizrr*, *verticalrr*, and *hybridrr* classify retention structure.
- *spread* is the issue spread to benchmark.
- *origsubpct* measures subordination.
- AAA, AA, A, BBB, and BB measure broad issue ratings.

- *horizrr_c* measures the continuous horizontal share of the required 5% retention.
- *anyloss*, *lossoverten*, and *lossrate_face* are used in the counterfactual loss analysis.

Why it matters. The paper relies on detailed hand-collected and vendor data. Table 1 clarifies exactly how the key retention and pricing variables are constructed.

6.3 Table 2: Security-level summary statistics

Read: This table describes the 2017–2019 security-level sample.

Key numbers:

- *horizrr* mean: **0.45**; *verticalrr* mean: **0.40**; *hybridrr* mean: **0.15**.
- Average spread: **1.38 percentage points**.
- About **38%** of rated securities are AAA.
- Average deal size: about **\$0.73 billion**.
- Average number of tranches: about **13.34**.
- Average horizontal retention size: **7.24% of par**.
- In pure horizontal deals, average horizontal retention is **7.92% of par** and median is **9.5% of par**.
- In pure vertical deals, vertical retention is exactly **5% of par**.

Interpretation. Horizontal retention has a much larger par footprint than vertical retention because the first-loss position is discounted relative to par.

Table 2
Security-level summary statistics, 2017–2019 vintage

	N	Mean	Median	SD	Min	Max
<i>horizrr</i>	3,332	.45	0	.5	0	1
<i>hybridrr</i>	3,332	.15	0	.36	0	1
<i>verticalrr</i>	3,332	.4	0	.49	0	1
<i>spread</i>	1,809	1.38	1.15	0.84	0.22	4.25
<i>firstcoupon</i>	2,815	4.01	3.98	1.05	1.94	8.45
AAA	2,242	.38	0	.49	0	1
AA	2,242	.12	0	.32	0	1
A	2,242	.12	0	.32	0	1
BBB	2,242	.15	0	.36	0	1
BB	2,242	.12	0	.33	0	1
nonAAAsr	1,729	.5	1	.5	0	1
<i>conduit</i>	3,332	.55	1	.5	0	1
<i>origsubpct</i>	2,548	22.72	21.61	17.87	0	70.92
<i>floaters</i>	3,261	.3	0	.46	0	1
<i>cutoffbalance</i>	3,292	0.73	0.76	0.34	0.11	1.56
<i>leadmgsz</i>	3,380	2.46	2.11	1.36	0.2	5.91
<i>depository</i>	3,380	.49	0	.5	0	1
<i>ntranches</i>	3,379	13.34	14	5.08	5	29
<i>mitg_orig_val</i>	2,840	6.72	7.39	3.41	1.8	10.04
<i>orig_wam</i>	3,054	83.31	110	40.26	22	120
<i>cutoffdisc_noi</i>	2,404	2.48	2.27	0.7	1.55	5.19
<i>cutoffltv</i>	3,054	57.35	57.77	7.65	32.3	77.3
<i>secwac</i>	2,144	4.52	4.58	0.45	2.77	5.37
<i>closeyear</i>	3,380	2017.78	2018	0.74	2017	2019
<i>horizcdfpar</i>	1,983	7.24	6.41	2.57	0.94	10.88
<i>vertize</i>	1,817	4.4	5	1.06	1.1	5
<i>horsize_hrr</i>	1,470	7.92	9.5	2.44	5	10.88
<i>horsize_hyrr</i>	513	5.27	5.66	1.83	0.94	8.34
<i>vertize_vrr</i>	1,299	5	5	0	5	5
<i>vertize_hyrr</i>	513	2.86	2.7	0.83	1.1	4.22
<i>tranchebalance</i>	3,054	61.97	33.7	79.52	0	415.47
<i>AAA_sub</i>	2,181	13.52	0	18.66	0	70.92
<i>AA_sub</i>	2,181	3.86	0	12	0	56.57
<i>A_sub</i>	2,181	2.78	0	9.22	0	47.06
<i>BBB_sub</i>	2,181	2.23	0	6.83	0	37.07
<i>horizrr_c</i>	3,314	0.52	0.63	0.47	0	1

Data cover all nonagency conduit and large loan/single-borrower U.S. CMBS issued between January 1, 2017, and June 30, 2019. Data come from Trepp, Bloomberg, and CRE Direct. Table 1 defines all variables. All variables are winsorized at the 1% level at both tails.

Table 3

Risk retention amount	Horizontal	Vertical	Hybrid	Retention	Vertical	Retention	Vertical	Retention	Hybrid	Retention	
ARMBS 2018-B01	5.00%	ARM 2017-PR02	5.00%	ARMBS 2017-GL03	5.00%	HYT 2019-AIT	5.00%	ARMBS 2018-C005	5.00%	CD 2017-C01	5.00%
ARMBS 2018-B02	5.00%	ARMBS 2017-B0K3	5.00%	ARM 2017-CJ0P	5.00%	CAMP 2018-L0P	5.00%	ARM 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B03	5.00%	ARMBS 2017-C1	5.00%	ARMBS 2017-GL03	5.00%	ARM 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B04	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B05	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B06	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B07	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B08	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B09	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B10	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B11	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B12	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B13	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B14	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B15	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B16	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B17	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B18	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B19	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B20	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B21	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B22	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B23	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B24	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B25	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B26	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B27	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B28	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B29	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B30	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B31	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B32	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B33	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B34	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B35	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B36	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B37	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B38	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B39	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B40	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B41	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B42	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B43	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B44	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B45	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B46	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B47	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B48	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B49	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B50	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B51	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B52	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B53	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%
ARMBS 2018-B54	5.00%	ARMBS 2017-B0K3	5.00%	ARMBS 2017-GL03	5.00%	ARMBS 2018-L0P	5.00%	ARMBS 2018-AP	5.00%	COMI 2017-C03	5.00%

6.4 Table 3: Risk retention amount

Read: The table lists retention amounts for deals where the authors can observe the necessary information.

Main takeaway:

- Vertical retention is exactly 5%.
- Horizontal and hybrid retention are almost exactly 5% of market value, with only tiny deviations above the minimum.

Interpretation. The rule binds. Issuers do not use the amount retained as a voluntary signal.

6.5 Figure 2: Cumulative losses and B-piece size, 2000–2016 conduit CMBS

Read: The left panel shows cumulative principal losses by vintage. The right panel shows B-piece size by vintage.

Key patterns:

- Cumulative losses are below 10% of original balance for all vintages except 2008.
- B-piece sizes before the rule are generally small relative to a 10% par threshold.

Why it matters. Since 5% of market value roughly maps to 10% of par for first-loss pieces, the figure shows that the mandated retention level is stringent relative to historical experience.

6.6 Figure 3: Density of B-piece size

Read: The figure plots the distribution of B-piece size for 2000–2016 conduit deals.

Main takeaway:

- The B-piece exceeds 10% of par in less than 1% of pre-rule conduit deals.

Interpretation. The post-rule horizontal requirement is not just slightly above the old norm. It is large relative to the historical distribution of first-loss pieces.

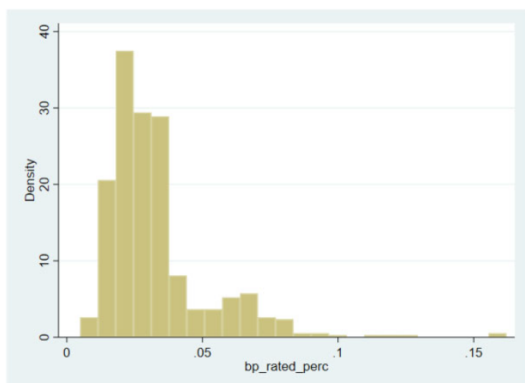


Figure 3
Density of B-piece size for 2000–2016 vintage conduit CMBS
This figure plots the distribution of the average BBB- subordination level (as a percentage of original deal level) for conduit CMBS.

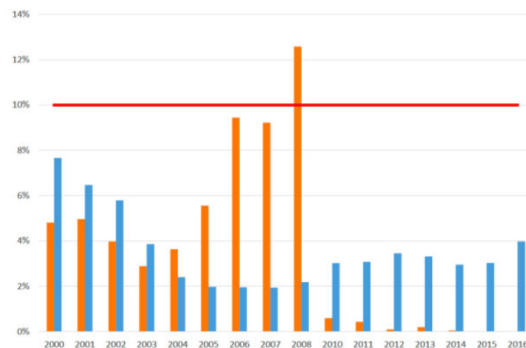


Figure 4
B-piece size versus cumulative losses for 2000–2016 vintage conduit CMBS
This figure overlays cumulative principal losses as a percentage of original deal balance (as of December 2018) with an average BBB- subordination level for conduit CMBS. Additionally, it displays a line for 10% of par value, which roughly corresponds to 5% of the market value based on estimates from the 2017–2019 data.

6.7 Figure 4: B-piece size versus cumulative losses

Read: The figure overlays historical B-piece sizes and cumulative losses and adds a 10% par-value line.

- Losses wipe out historical B-pieces for 2004–2008 vintages.
- However, losses exceed the 10% par threshold only for the 2008 vintage.

Interpretation. A 5% market-value horizontal retention rule would have protected senior securities in almost all historical vintage years. The rule is therefore stringent in ordinary market conditions.

6.8 Figure 5: Issuer payoffs with horizontal and vertical retention

Read: The figure plots issuer payoffs $\pi_h(\beta)$ and $\pi_v(\beta)$ from the numerical model example.

Key numbers:

- $\hat{\beta} = 0.00438$.
- $h = 0.0996$.
- $P_v = 0.8954$ and $P_h = 0.9004$.
- Issuers with $\beta \leq 0.00438$ choose horizontal retention; issuers with higher β choose vertical retention.

Interpretation. Horizontal retention is more costly for bad collateral pools, so it separates better pools from worse pools.

6.9 Table 4: Yield spreads and the form of risk retention

Read: This is the paper’s central pricing table. It estimates the baseline spread regression.

Key numbers:

- The coefficient on *horizrr* ranges from **-0.059** to **-0.110** percentage points.
- In basis points, this is about **5.9 to 11 bp** lower issue spreads for horizontal deals.
- The result is statistically significant across specifications.
- The average spread is about 138 bp, so the effect is economically meaningful.

Interpretation. Investors price horizontal retention as good news about collateral quality. This is exactly the model’s prediction.

6.10 Table 5: Heterogeneous effects of risk retention on yield spreads

Read: This table interacts horizontal retention with rating indicators.

Key numbers:

- The interaction *BBB_h* is negative and significant across specifications.

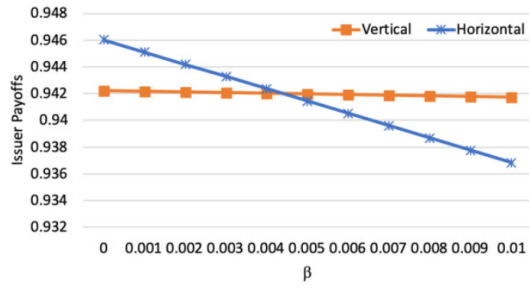


Figure 5
Issuer payoffs with horizontal and vertical retention
This figure shows issuer payoffs as a function of β , conditional on $v=0.05$ (required level of risk retention), $\beta_0=0.005$ (mean of β), $\alpha_0=0.05$ (mean of α), $\delta_v=0.98$ (issuer discount factor with vertical retention), and $\delta_h=0.92$ (issuer discount factor with horizontal retention). v is the mandatory horizontal risk retention levels. β , plotted along the x-axis, represents the portion of losses on a pool of mortgages not predictable to outside investors based on observable characteristics. The issuer observes the exact value of β when it chooses the form of risk retention; outside investors know only the distribution of β .

Table 4
Yield spreads and the form of risk retention

	(1)	(2)	(3)	(4)	(5)
<i>horizrr</i>	-0.059* (0.030)	-0.062** (0.030)	-0.066** (0.033)	-0.080*** (0.029)	-0.11** (0.043)
<i>hybridrr</i>	-0.051 (0.032)	-0.054* (0.032)	-0.029 (0.035)	-0.080** (0.038)	-0.059 (0.050)
<i>AAA</i>	-2.91*** (0.17)	-1.89*** (0.069)	-1.86*** (0.070)	-2.13*** (0.082)	-2.33*** (0.099)
<i>AA</i>	-2.43*** (0.16)	-1.41*** (0.059)	-1.38*** (0.057)	-1.73*** (0.076)	-1.90*** (0.084)
<i>A</i>	-2.03*** (0.15)	-1.01*** (0.049)	-0.97*** (0.044)	-1.23*** (0.071)	-1.36*** (0.080)
<i>BBB</i>	-1.01*** (0.16)				
<i>BB</i>	-0.69*** (0.11)				
<i>conduit</i>	0.51*** (0.088)	0.51*** (0.090)	0.50*** (0.090)	0.63*** (0.11)	0.64*** (0.20)
<i>origsubpct</i>	0.0063*** (0.0016)	0.0059*** (0.0017)	0.0059*** (0.0017)	0.0099*** (0.0026)	0.015*** (0.0041)
<i>tranchebalance</i>				-0.00082*** (0.000100)	-0.00094*** (0.00012)
<i>floaterr</i>	0.0013 (0.11)	0.046 (0.12)	0.14 (0.086)	0.17* (0.089)	0.024 (0.23)
<i>cutoffbalance</i>	0.017 (0.050)	0.0031 (0.049)	0.054 (0.051)	0.030 (0.052)	0.12 (0.091)
<i>leadmgrsize</i>	-5.5e-06 (4.0e-06)	-5.9e-06 (4.0e-06)	0.000011 (7.3e-06)	0.000011 (7.0e-06)	0.000015* (8.3e-06)
<i>depository</i>	-0.0033 (0.027)	-0.012 (0.026)			
<i>ntranches</i>	-0.0044 (0.0042)	-0.0038 (0.0042)	-0.0050 (0.0045)	-0.0091* (0.0053)	-0.0012 (0.0076)
<i>mtg_orig_val</i>	0.094*** (0.0062)	0.098*** (0.0063)	0.10*** (0.0056)	0.13*** (0.0079)	0.12*** (0.0095)
<i>orig_wam</i>	-0.0087*** (0.0012)	-0.0085*** (0.0012)	-0.0082*** (0.0012)	-0.0074*** (0.0010)	-0.0079*** (0.0019)
<i>cutoffdscr_noi</i>				0.13*** (0.043)	-0.21** (0.100)
<i>cutoffltv</i>				-0.00013 (0.0040)	-0.0008 (0.0076)
<i>secwac</i>				0.20*** (0.064)	0.27** (0.11)
<i>firstcoupon</i>	0.054 (0.034)	0.029 (0.035)	0.0022 (0.031)	-0.031 (0.036)	0.032 (0.042)
Observations	1,557	1,500	1,467	1,027	749
<i>R</i> ²	.836	.820	.828	.901	.914
Year-quarter FEs	Yes	Yes	Yes	Yes	Yes
Lead manager FEs	No	No	Yes	Yes	Yes
Bpcc buyer FEs	No	No	No	No	Yes
Property type share controls	Yes	Yes	Yes	Yes	Yes

This table provides the results of estimating linear regressions of initial tranche pricing on risk retention indicators and controls. The dependent variable is the spread to benchmark at issue in percentage points. We do not show the constant term. Column 1 includes all securities in the sample, and Columns 2-5 include only securities with an initial rating of BBB- or higher. Data cover nonagency conduit and large loan/single-borrower U.S. CMBS issued between January 1, 2017, and June 30, 2019. Table 1 defines all variables, which are winsorized at the 1% level at both tails. *** $p < .01$; ** $p < .05$; * $p < .1$. Standard errors are clustered at the deal level.

- BBB securities in horizontal deals have spreads about **37 to 75 bp** lower than comparable BBB securities in non-horizontal deals.
- The interaction *nonAAAsr_h* is also negative, with magnitudes of about **10 to 17 bp**.

Interpretation. The horizontal signal matters most for tranches that are closer to expected losses. AAA securities are less sensitive because they are more protected.

Table 5
Heterogeneous effects of risk retention on yield spreads

	(1)	(2)	(3)	(4)	(5)	(6)
<i>horizrr</i>	0.031 (0.030)	0.031 (0.034)	-0.031 (0.029)	-0.071* (0.039)	-0.024 (0.042)	0.0036 (0.056)
<i>hybridrr</i>	-0.056 (0.034)	-0.056 (0.043)	-0.081** (0.036)	-0.049 (0.043)	-0.035 (0.045)	0.0089 (0.056)
<i>nonAAAsr</i>		0.75*** (0.046)		0.68*** (0.055)		0.56*** (0.10)
<i>nonAAAsr_h</i>		-0.10* (0.052)		-0.11 (0.067)		-0.17** (0.080)
<i>BBB</i>	1.41*** (0.081)		1.76*** (0.098)		1.91*** (0.10)	
<i>BBB_h</i>	-0.37*** (0.11)		-0.54*** (0.13)		-0.75*** (0.16)	
<i>conduit</i>	-0.016 (0.087)	-0.20 (0.13)	-0.073 (0.10)	-0.73*** (0.15)	-0.39* (0.21)	-1.52*** (0.33)
<i>origsubpct</i>	-0.010*** (0.0016)	-0.018*** (0.0024)	-0.0094*** (0.0028)	-0.025*** (0.0038)	-0.014** (0.0055)	-0.040*** (0.0084)
<i>tranchebalance</i>			-0.0016*** (0.00019)	-0.00095*** (0.00015)	-0.0014*** (0.00031)	-0.00055** (0.00024)
<i>floaters</i>	0.092 (0.13)	-0.011 (0.20)	0.064 (0.18)	-0.26* (0.15)	0.28 (0.23)	-0.21 (0.26)
<i>cutoffbalance</i>	-0.029 (0.053)	-0.048 (0.071)	0.099* (0.056)	0.10 (0.082)	0.076 (0.099)	0.019 (0.12)
<i>leadmgrsize</i>	-2.3e-06 (4.0e-06)	-1.0e-07 (5.2e-06)	-2.4e-07 (3.3e-06)	-3.7e-06 (4.6e-06)	-7.7e-06* (4.3e-06)	-0.000014** (6.1e-06)
<i>depository</i>	-0.022 (0.026)	-0.044 (0.034)	-0.011 (0.025)	0.00033 (0.032)	0.00025 (0.033)	0.041 (0.040)
<i>ntranches</i>	0.0058 (0.0059)	0.018** (0.0088)	-0.0054 (0.0040)	0.00078 (0.0049)	0.0014 (0.0061)	0.0066 (0.0071)
<i>mtg_orig_wal</i>	0.11*** (0.0061)	0.10*** (0.0083)	0.14*** (0.011)	0.18*** (0.0094)	0.12*** (0.015)	0.16*** (0.014)
<i>orig_wam</i>	-0.010*** (0.0014)	-0.0088*** (0.0021)	-0.012*** (0.0016)	-0.014*** (0.0017)	-0.0079*** (0.0023)	-0.0084*** (0.0026)
<i>cutoffdscr_noi</i>			0.063 (0.047)	-0.014 (0.072)	0.0057 (0.095)	0.23* (0.12)
<i>cutoffltv</i>			0.010** (0.0044)	0.013*** (0.0050)	0.017* (0.0085)	0.035*** (0.012)
<i>secvac</i>			0.23*** (0.063)	0.55*** (0.098)	0.26*** (0.082)	0.52*** (0.098)
<i>firstcoupon</i>	0.099*** (0.031)	-0.089* (0.052)	0.084** (0.036)	-0.47*** (0.036)	0.15*** (0.038)	-0.44*** (0.043)
Observations	1,500	1,500	1,049	1,049	769	769
R ²	.742	.625	.862	.748	.885	.770
Year-quarter FEs	Yes	Yes	Yes	Yes	Yes	Yes
Bpce buyer FEs	No	No	No	No	Yes	Yes
Property type share controls	Yes	Yes	Yes	Yes	Yes	Yes

Results of estimating linear regressions of initial tranche pricing on risk retention indicators, interactions between retention indicators and rating indicators, and controls. The dependent variable is the spread to benchmark at issue in percentage points. We do not show the constant term. Data cover senior securities (BBB- and above) only. Data cover nonagency conduit and large loan/single-borrower U.S. CMBS issued between January 1, 2017, and June 30, 2019. Table 1 defines all variables, which are winsorized at the 1% level at both tails. *** $p < .01$; ** $p < .05$; * $p < .1$. Standard errors are clustered at the deal level.

Table 6
Robustness

	(1)	(2)	(3)	(4)	(5)	(6)
<i>horizrr</i>	-0.092** (0.043)					
<i>hybridrr</i>	-0.051 (0.050)					
<i>horizrr_c</i>		-0.053* (0.030)	-0.055* (0.030)	-0.063* (0.032)	-0.061** (0.030)	-0.081* (0.044)
<i>AAA_sub</i>	0.017 (0.012)					
<i>AA_sub</i>	0.020 (0.013)					
<i>A_sub</i>	0.014 (0.014)					
<i>BBB_sub</i>	-0.027 (0.018)					
Observations	749	1,557	1,500	1,467	1,027	749
R ²	.915	.836	.820	.828	.901	.914
Year-quarter FEs	Yes	Yes	Yes	Yes	Yes	Yes
Baseline controls	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	Yes	No	No	No	Yes	Yes
Lead manager FEs	Yes	No	No	No	Yes	Yes
Bpce buyer FEs	Yes	No	No	No	No	Yes
Property type share controls	Yes	Yes	Yes	Yes	Yes	Yes

This table provides the results of estimating linear regressions of initial tranche pricing on risk retention indicators and controls. The dependent variable is the spread to benchmark at issue in percentage points. We do not show the constant term. Data cover nonagency conduit and large loan/single-borrower U.S. CMBS issued between January 1, 2017, and June 30, 2019. Columns 1 and 3–6 are estimated on senior securities (BBB- and above) only, and Column 2 is estimated on all securities. “Baseline controls” comprise the controls included in Columns 1–3 of Table 4, and “Additional controls” are *tranchebalance*, *cutoffdscr_noi*, *cutoffltv*, and *secvac*. Table 1 defines all variables, which are winsorized at the 1% level at both tails. *** $p < .01$; ** $p < .05$; * $p < .1$. Standard errors are clustered at the deal level.

6.11 Table 6: Robustness

Read: The table reports two robustness exercises.

- Column 1 adds rating-by-subordination controls to allow a nonlinear relation between subordination and spreads. The horizontal coefficient remains negative and significant, about **-9.2 bp**.
- Columns 2–6 replace the horizontal dummy with *horizrr_c*, a continuous measure of how horizontal the retention is. Coefficients range from about **-5.3 to -8.1 bp** and are significant.

Interpretation. The spread result is not driven only by coarse horizontal-versus-non-horizontal classification. More horizontal exposure is associated with lower spreads.

Table C.1
Loan-level summary statistics

A. 2006–2008 vintages						
	N	Mean	Median	SD	Min	Max
<i>lossrate_face</i>	23,325	.09	0	.22	0	1
<i>face</i>	23,315	17.42	6.32	66.45	0.08	7,407.65
<i>securlv</i>	23,183	68.65	71.7	11.9	1.3	102
<i>orig_dscrmoi</i>	22,158	1.67	1.49	2.47	-0.5	204.17
<i>securwac</i>	23,225	6	5.95	0.39	1.21	11.81
<i>securroc</i>	22,150	92.75	96.9	10.15	2.3	100
<i>term_secur</i>	23,062	8.86	9.73	3.21	0.06	34.83
<i>origyear</i>	23,325	2,006.46	2006	0.51	2,006	2,008
<i>anyloss</i>	23,325	0.20	0	0.39	0	1
<i>losssoverten</i>	23,325	0.16	0	0.37	0	1
B. 2017–2019 vintage						
<i>lossrate_face</i>	5,334	0	0	0	0	0
<i>face</i>	5,331	32.35	12.7	97.15	0	1,880
<i>securlv</i>	5,247	59.4	62.7	13.7	1	94.28
<i>orig_dscrmoi</i>	5,005	2.31	1.84	2.43	0.74	57.46
<i>securwac</i>	4,762	4.86	4.86	0.66	2	10.39
<i>securroc</i>	5,134	91.03	94	9.6	24.2	100
<i>term_secur</i>	5,240	8.77	9.88	2.81	0.38	24.98
<i>origyear</i>	5,334	2,017.71	2,018	0.71	2,017	2,019

Data cover all 2006, 2007, 2008, 2017, 2018, and January–June 2019 vintage nonagency conduit and large loan/single-borrower U.S. CMBS loans. Data come from Trepp. Table 1 defines all variables.

Table C.2
Loan-level characteristics: *t*-tests

	2006–2008 vintage	2017–2019 vintage	<i>t</i> -stat	<i>p</i> -value
<i>face</i>	17.42	32.35	−10.66	0.00
<i>securlv</i>	68.65	59.4	45.16	0.00
<i>orig_dscrmoi</i>	1.67	2.31	−16.78	0.00
<i>securwac</i>	6	4.86	115.14	0.00
<i>securroc</i>	92.75	91.03	11.48	0.00
<i>term_secur</i>	8.86	8.77	2.01	0.04
<i>origyear</i>	2,006.46	2,017.71	−1,093.81	0.00

This table provides the results of comparing means across loan-level variables. Columns 4 and 5 report *t*-stats and *p*-values. Data cover all U.S. nonagency conduit and large loan/single-borrower CMBS loans issued between January 2006 and December 2008 (Column 2) or between January 2017 and June 2019 (Column 3). Data come from Trepp. Table 1 defines all variables.

6.12 Appendix C tables: counterfactual loss analysis

Read: Appendix C asks how 2017–2019 loans would have performed under crisis-era conditions.

Key numbers:

- 2006–2008 loans had actual mean loss of about **9%** of face value.
- 2017–2019 loans are predicted to have mean losses of about **5.5% to 6.2%** if mapped through crisis-era loss models.
- Actual any-loss probability in 2006–2008 is about **19.5%**; predicted any-loss probability for 2017–2019 loans is roughly **13%**.
- Actual probability of loss above 10% in 2006–2008 is about **16.2%**; predicted probability for 2017–2019 loans is roughly **10%**.

Interpretation. Post-rule collateral quality appears safer than crisis-era collateral. This reinforces the claim that the mandated retention level is stringent.

Table C.3
Predictor loan loss indicator regressions: 2006–2008 vintages

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	<i>anyloss</i>	<i>anyloss</i>	<i>anyloss</i>	<i>anyloss</i>	<i>losssoverten</i>	<i>losssoverten</i>	<i>losssoverten</i>	<i>losssoverten</i>
<i>face</i>	0.000070 (0.00015)	0.000070 (0.00015)	−0.00026 (0.00030)	−0.00067 (0.00045)				
<i>securlv</i>	0.028*** (0.0016)	0.028*** (0.0015)	0.029*** (0.0016)	0.030*** (0.0015)	0.031*** (0.0015)	0.031*** (0.0015)	0.031*** (0.0016)	0.032*** (0.0017)
<i>orig_dscrmoi</i>	−0.015* (0.0087)	−0.015* (0.0088)	−0.016 (0.0088)	−0.018 (0.0092)	−0.019 (0.012)	−0.019 (0.012)	−0.025 (0.017)	−0.028* (0.017)
<i>securroc</i>			−0.0088*** (0.0014)	−0.0072*** (0.0017)	−0.0072*** (0.0017)	−0.0072*** (0.0017)	−0.0072*** (0.0017)	−0.0072*** (0.0017)
<i>term_secur</i>			−0.002*** (0.0073)	−0.002*** (0.0073)	−0.002*** (0.0073)	−0.002*** (0.0073)	−0.002*** (0.0073)	−0.002*** (0.0073)
<i>conduit</i>			0.42* (0.21)	0.42* (0.21)	0.42* (0.21)	0.42* (0.21)	0.42* (0.21)	0.42* (0.21)
<i>Constant</i>	−2.80*** (0.12)	−2.70*** (0.12)	−2.40*** (0.25)	−2.40*** (0.25)	−3.13*** (0.12)	−3.13*** (0.12)	−2.51*** (0.25)	−2.51*** (0.30)
Observations	22,140	22,140	20,929	20,885	22,148	22,148	20,937	20,861
Pseudo <i>R</i> ²	.034	.035	.077	.084	.039	.040	.095	.085
Mean pred. prob., 2017–2019Q2	0.136	0.136	0.132	0.131	0.106	0.108	0.104	0.102
Actual prob., 2006–2008	0.195	0.195	0.195	0.195	0.162	0.162	0.162	0.162

Results of estimating probit regressions of realized loan loss indicator variables on controls. The “Mean pred. prob.” is the average predicted probability of the outcome based on the coefficient estimates, and the “Actual prob.” is the sample frequency of the outcome. Data cover loans originated and securitized in emergency conduit and large loan/single-borrower U.S. CMBS deals between January 2006 and December 2008. Table 1 defines all variables. ****p* < .01; ***p* < .05; **p* < .1. Standard errors are clustered at the deal level.

Table C.4
Predictor loan loss size regressions: 2006–2008 vintages

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	<i>Censored</i>	<i>Censored</i>	<i>Censored</i>	<i>Censored</i>	<i>OLS</i>	<i>OLS</i>	<i>OLS</i>	<i>OLS</i>
<i>securlv</i>	0.027*** (0.0013)	0.027*** (0.0013)	0.028*** (0.0013)	0.029*** (0.0017)	0.028*** (0.00017)	0.028*** (0.00015)	0.028*** (0.00015)	0.027*** (0.00016)
<i>orig_dscrmoi</i>	−0.0086 (0.0069)	−0.0087 (0.0070)	−0.0092 (0.014)	−0.012 (0.013)	−3.1e-06 (0.00071)	−3.6e-06 (0.00071)	−0.00067 (0.0016)	−0.0008 (0.0016)
<i>securroc</i>			−0.0088*** (0.0012)	−0.0088*** (0.0012)	−0.0088*** (0.0012)	−0.0088*** (0.0012)	−0.0088*** (0.0012)	−0.0088*** (0.0012)
<i>term_secur</i>			−0.0020 (0.0073)	−0.0020 (0.0073)	−0.0020 (0.0073)	−0.0020 (0.0073)	−0.0020 (0.0073)	−0.0020 (0.0073)
<i>conduit</i>			0.20 (0.17)	0.25 (0.16)	0.25 (0.16)	0.25 (0.16)	0.25 (0.16)	0.25 (0.16)
<i>Constant</i>	−3.27*** (0.097)	−3.27*** (0.097)	−3.27*** (0.22)	−4.89*** (0.37)	−0.11*** (0.012)	−0.11*** (0.012)	−0.042* (0.023)	−0.12*** (0.028)
Observations	22,148	22,148	20,937	20,909	22,148	22,148	20,937	20,869
Pseudo <i>R</i> ²	.031	.032	.032	.079	.024	.024	.026	.076
Predicted mean loss, 2017–2019Q2	0.038	0.038	0.037	0.037	0.062	0.062	0.058	0.055
Actual mean loss, 2006–2008	0.09	0.09	0.09	0.09	0.09	0.09	0.09	0.09

Results of estimating fractional response regressions (Columns 1–4) and OLS regressions (Columns 5–8) of realized loan losses (*lossrate_face*) on controls. The “Predicted mean loss” is the average predicted loss (as a percentage of par value) based on the coefficient estimates, and the “Actual mean loss” is the mean realized loss (as a percentage of par value) in the sample. Data cover loans originated and securitized in nonagency conduit and large loan/single-borrower U.S. CMBS deals between January 2006 and December 2008. Table 1 defines all variables. ****p* < .01; ***p* < .05; **p* < .1. Standard errors are clustered at the deal level.

7 Policy implications

7.1 Investor sophistication

The paper’s pricing evidence implies that CMBS investors understand the signaling role of retention structure. They price horizontally retained deals more favorably, especially for more information-sensitive tranches.

Interpretation. This is not a market in which investors simply ignore issuer incentives. Post-crisis CMBS investors appear sophisticated enough to infer quality from the retention structure.

7.2 The rule may reduce informational efficiency

The mandatory 5% level can protect investors, but it also removes information.

- Without a binding rule, issuers might signal quality by choosing different levels of retention.
- With the binding 5% rule, issuers pool at the minimum level.
- The remaining signal is retention structure, which is coarser than level-based signaling.

Interpretation. A partially separating equilibrium is better than complete pooling, but it does not reveal exact asset quality. Investors learn whether collateral quality is above or below a threshold, not the full type.

7.3 Two positive design features

The authors view two parts of the rule as clearly useful:

- **Five-year holding period.** The B-piece buyer cannot immediately resell the first-loss position, so horizontal retention represents genuine exposure.
- **Market-value measurement.** Requiring horizontal retention to be measured by market value rather than par value prevents low-quality deals from satisfying the rule with cheap, risky first-loss par.

Interpretation. The problem is not necessarily risk retention itself. The concern is whether the required level is too high for normal market conditions.

7.4 Dynamic retention policy

The paper suggests that a dynamic or state-contingent retention requirement may be better than a fixed 5% rule.

- In crisis-like environments with weak underwriting, high required retention may be useful.
- In safer post-crisis markets with sophisticated investors, the same level may be unnecessarily stringent.
- If collateral quality deteriorates, asymmetric information may rise again, so the optimal rule may change over time.

Policy takeaway. The right retention rule depends on market conditions, investor sophistication, collateral quality, and the degree of asymmetric information.

8 Short summary

- **What the paper studies.** Dodd–Frank risk retention in post-2016 CMBS and its effect on the information investors infer from retention choices.
- **Core institutional fact.** The 5% retention requirement binds in all observable cases and is stringent relative to historical CMBS losses.
- **Main theory.** Since issuers cannot signal using retention levels, they may signal using retention structure. Horizontal first-loss retention is costlier for low-quality collateral, so high-quality issuers choose it.
- **Main empirical result.** Horizontal retention is associated with about 5.9–11 bp lower issue spreads. The effect is strongest for BBB and other non-AAA senior tranches.
- **Quantitative implication.** The implied asymmetric information in post-crisis CMBS is small, roughly 0.7–1.3% of deal value.
- **Policy lesson.** Risk retention can improve incentives, but a fixed high requirement can also reduce informational efficiency by eliminating level-based signaling.

9 Conclusion

Core conclusion. Flynn, Ghent, and Tchistyi show that post-Dodd–Frank CMBS issuers do not use the level of risk retention to signal quality because the mandated 5% retention requirement binds. However, issuers can still signal quality through retention structure. Pure horizontal first-loss retention is associated with better pricing, especially for information-sensitive tranches.

Economic intuition. Horizontal retention exposes the issuer or B-piece buyer to the first losses in the deal. This exposure is costly when collateral quality is poor and less costly when collateral quality is good. Therefore, choosing horizontal retention credibly communicates confidence in the collateral pool.

Takeaway. Regulation changes both incentives and information. A high minimum retention rule can protect investors, but it may also remove useful market signals. In this paper’s post-crisis CMBS market, investors appear sophisticated, asymmetric information is low, and the fixed 5% rule may be more stringent than necessary under normal market conditions.